

$$\text{Zam } \Rightarrow \frac{1}{1-x} = \sum_{k=0}^{\infty} x^k \quad x \in (-1, 1)$$

$$x \leftrightarrow -x \Rightarrow \frac{1}{1+x} = \sum_{k=0}^{\infty} (-1)^k x^k, \quad x \in (-1, 1)$$

$$\Rightarrow \ln(1+x) = \sum_{k=0}^{\infty} (-1)^k \int_0^x x^k dx =$$

$$= \sum_{k=0}^{\infty} (-1)^k \frac{x^{k+1}}{k+1} = \sum_{k=1}^{\infty} (-1)^{k-1} \frac{x^k}{k}$$

$$\textcircled{\text{T}} \quad \operatorname{arctg} x = \sum_{k=1}^{\infty} (-1)^{k-1} \frac{x^{2k-1}}{2k-1}, \quad x \in [-1, 1]$$

$$\Delta \quad \frac{1}{1+x^2} = \sum_{k=0}^{\infty} (-1)^k x^{2k}, \quad x \in (-1, 1)$$

$$\operatorname{arctg} x = \sum_{k=1}^{\infty} (-1)^{k-1} \frac{x^{2k-1}}{2k-1},$$

$$\textcircled{\text{T}} \quad x = \pm 1 \Rightarrow \text{при } x \text{ не } \Pi p \text{ д-во}$$

$$\Rightarrow \text{при } x \text{ не } [-1, 1]$$

$$\textcircled{\text{T}} \quad (1+x)^\alpha = 1 + \sum_{k=1}^{\infty} \frac{\alpha(\alpha-1)\dots(\alpha-k+1)}{k!} x^k$$

$$x \in (-1, 1), \quad \alpha \in \mathbb{R}$$

$$\textcircled{\text{T}} \quad \arcsin x = x + \sum_{k=1}^{\infty} \frac{(2k-1)!!}{(2k)!!} \frac{x^{2k+1}}{2k+1}$$

$$x \in [-1, 1]$$

$$\star \quad (1-x^2)^{-1/2} = \uparrow$$

Q_{np} $\mathbb{P}_{eg} \quad \frac{a_0}{2} + \sum_{k=1}^{\infty} a_k \cos kx + b_k \sin kx$

каж. тригоном - и функц, постр - и
по φ -и $\{1, \cos kx, \sin kx\}, k \in \mathbb{N}$

$$\exists k, m \in \mathbb{N} \Rightarrow$$

$$\begin{aligned} \textcircled{1} \quad \int_{-\pi}^{\pi} \sin kx \cdot \cos mx \, dx &= \int_{-\pi}^{\pi} 1 \cdot \cos kx \, dx = \int_{-\pi}^{\pi} 1 \cdot \sin kx \, dx = 0 \\ \int_{-\pi}^{\pi} \sin kx \cdot \sin mx \, dx &= \int_{-\pi}^{\pi} \cos kx \cdot \cos mx \, dx = 0 \end{aligned}$$

$$\textcircled{2} \int_{-\pi}^{\pi} \sin^2 x \, dx = \int_{-\pi}^{\pi} \cos^2 x \, dx = \pi$$
$$\int_{-\pi}^{\pi} 1^2 \, dx = 2\pi$$

③ $f(x) : f(x + 2\pi) = f(x), \quad T = 2\pi$

$$f(x) = \frac{a_0}{2} + \sum_{k=1}^{\infty} a_k \cos kx + b_k \sin kx \quad (*)$$

$\int_{-\pi}^{\pi} f(x) dx = 0$
 и $f(x) \in C^0[-\pi, \pi]$

9 $\int_{-\pi}^{\pi} f dx = \frac{1}{2} \int_{-\pi}^{\pi} a_0 dx = \frac{1}{2} a_0 \cdot 2\pi = \pi a_0$

$$\Rightarrow a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f \, dx$$

$$\int_{-\pi}^{\pi} f(x) \cos mx \, dx = \frac{a_0}{2} \int_{-\pi}^{\pi} \cos mx \, dx + \sum_{k=1}^{\infty} (a_k \int_{-\pi}^{\pi} \overset{=0}{\cos kx} \cdot \overset{=0}{\cos mx} \, dx + b_k \int_{-\pi}^{\pi} \overset{=0}{\sin kx} \cdot \cos mx \, dx) = a_k \cdot \pi$$

$$a_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos kx \, dx, \quad k = 1, 2, \dots$$

$$b_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin kx \, dx, \quad k = 1, 2, \dots$$

Доп Если же $f \in L^1$ a_k, b_k — не ϕ -чл. $\uparrow$.

$$\frac{a_0}{2} + \sum_{k=1}^{\infty} a_k \cos kx + b_k \sin kx \text{ не сж.}$$

при этом периодич. Фурье ϕ -чл. f ,
а a_k, b_k — коэф. Фурье ϕ . f — не

сж. $\{1, \cos kx, \sin kx, k \in \mathbb{N}\}$. $\Rightarrow$

$$f(x) \sim \frac{a_0}{2} + \sum_{k=1}^{\infty} a_k \cos kx + b_k \sin kx$$

Тег Фурье в комплекс. форме

$$\begin{cases} e^{ikx} = \cos kx + i \sin kx, & k \in \mathbb{N} \\ e^{-ikx} = \cos kx - i \sin kx. \end{cases}$$

$$\cos kx = \frac{e^{ikx} + e^{-ikx}}{2} \quad \sin kx = \frac{e^{ikx} - e^{-ikx}}{2i}$$

$$\begin{aligned}
& \Rightarrow \frac{a_0}{2} + \sum_{k=1}^{\infty} a_k \frac{e^{ikx} + e^{-ikx}}{2} + b_k \frac{e^{ikx} - e^{-ikx}}{2i} = \\
& = \frac{a_0}{2} + \sum_{k=1}^{\infty} e^{ikx} \left[\frac{a_k - ib_k}{2} \right] + e^{-ikx} \left[\frac{a_k + ib_k}{2} \right] \\
& = \sum_{k=-\infty}^{+\infty} C_k e^{ikx}
\end{aligned}$$

$$C_k = \begin{cases} \frac{a_k - ib_k}{2}, & k > 0 \\ a_0/2, & k = 0 \\ \frac{a_{-k} + ib_{-k}}{2}, & k < 0 \end{cases}$$

$$\begin{aligned}
C_k &= \frac{1}{2} \left[\frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos kx \, dx - \frac{i}{\pi} \int_{-\pi}^{\pi} f(x) \sin kx \, dx \right] = \\
&= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) (\cos kx - i \sin kx) \, dx = \\
&= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-ikx} \, dx
\end{aligned}$$

$$C_k = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-ikx} \, dx \quad \text{for } \forall k$$

$$C_0 = \frac{a_0}{2} = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) \, dx.$$

$$\begin{aligned}
C_{-k} &= \frac{1}{2\pi} \left[\int_{-\pi}^{\pi} f(x) (\cos kx + i \sin kx) \, dx \right] = \\
&= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{ikx} \, dx.
\end{aligned}$$

Дип. Существует ли $\exists C_k$ $\forall k \in \mathbb{Z}$, то

$$\sum_{k=-\infty}^{\infty} C_k e^{ikx}$$

мож. представ Φ . в Φ форме

Φ - ун f , C_k - k -ум Φ Φ -ум f

отн. сист. Φ . $\{e^{ikx}, k \in \mathbb{Z}\}$

$$\begin{aligned} \int_{-\pi}^{\pi} f e^{-ikx} dx &= \int_{-\pi}^{\pi} f [\cos kx - i \sin kx] dx = \\ &= \int_{-\pi}^{\pi} f \cos kx dx - i \int_{-\pi}^{\pi} f \sin kx dx. \end{aligned}$$

$$T_n(x) = \sum_{k=-n}^n C_k e^{ikx} = \sum_{k=-n}^n \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} f e^{-ikt} dt \right) e^{ikx}$$

$$* e^{ikx} = \frac{1}{2\pi} \int_{-\pi}^{\pi} f \sum_{k=-n}^n e^{ik(x-t)} dt$$

$$\odot \text{ сформулируем } e^{ip} \neq 1 \Leftrightarrow p \neq 2\pi k$$